



THE NEXT GENERATION

6G can bring terabit per second speeds to your mobile device

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5G

technology is in the process of moving from a research phase to a deployment

phase. 3GPP is the standards organisation that has been developing the protocols for cellular networks. The protocols are updated via a series of releases, and the first phase of 5G is release 15. Before the releases, proposals are accepted for the technical specifications of the standards, and the protocols are updated according to the requirements. The 6G rollout is expected to start in the 2030s, by which time, the world will be a very different place.

We can expect level 5 autonomous cars, which are entirely driverless. These vehicles would have to talk to each other constantly. Parcels would move across roadways in autonomous

truck platoons, which would also have to talk to each other constantly for safe navigation and reacting rapidly to changing environments. Delivery by drones would be common, and at least in some cities, air taxis are expected to be a common sight. Pretty much every product would be part of the IoT. It would be very difficult for anything to get lost. Additionally, almost everything is expected to be infused with AI, which would require near constant connectivity. AR and VR technologies are expected to be ubiquitous and cheap. Researchers also expect advancements in volumetric or holographic displays. Autonomous and mobile robots would have become more common, and quantum computing is expected to go mainstream by then. Almost every activity that humans indulge in, is expected to be connected to the internet in some way. Flexible screens could be applied on

any surface, from packaged products to windows. Apart from personal electronics, a myriad of interfaces would be available in public and shared spaces, which would require the network itself to be trusted with biometric identification and security. Buildings, vehicles, cities, homes and the things within them would all be smart and constantly be chattering to each other. This would require an unprecedented volume of devices to be connected all the time, with urban areas seeing a density that is expected to increase exponentially. The rollout of 5G is designed to accommodate all these use cases, but these are only the applications that we can think of now.

If you went back in time to 2009, and asked even the best engineers about what would the demand on cellular networks be in 2019, they would not be able to come up with all the applications that we see now. While